Prashant Kumar Dubey

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
St. Mary's School	Class XII (ISC)	2019	82.50%
St. Mary's School	Class X (ICSE)	2017	80.40%

TECHNICAL SKILLS

Programming Languages: Java, C, SQL, Python

Deep Learning / ML: PyTorch, TensorFlow, Torchvision, NumPy, Pandas, scikit-learn, HuggingFace, Mat- plotlib

Tools: Git, GitHub, Kaggle, Jupyter, Google Colab

EXPERIENCE

Amazon ML Summer School 2026

Acknowledgement Letter

- Learned core ML topics including Supervised Learning, Deep Neural Networks, Dimensionality Reduction, Un-supervised Learning, Generative AI & LLMs, Sequential Learning, Causal Inferencing, and Reinforcement Learning.

PROJECTS

KLA AI Hack x IIT-H 2026 - Image Restoration & Super-Resolution 2026

- GitHub: pkd-prashant/KLA-AI-HACK-x-IIT-H-2026

- Built an end-to-end pipeline for joint denoising and 2× super-resolution of speckled

128×128 images to clean 256×256 outputs.

- Formulated and evaluated ResUNet/HAT-L transformer (Hybrid Attention Transformer) architectures, culminating in a final model that

secured an SSIM(Structural Similarity Index Measure) of 0.877.

- Designed a composite training objective combining **Charbonnier loss, MS-SSIM, and frequency-domain L1 loss** to improve reconstruction quality and preserve high-frequency pixels.

Synthetic 3D-Scene Classification

GitHub: pkd-prashant/IIT-H-Deep-Learning-2026-Hackathon

- Developed a dual-stream PyTorch classification pipeline using ResNet-34, combining RGB features with gradient-based features from Canny edges, Sobel magnitude, and phase to capture complementary visual and structural cues.

- Trained the model using Binary Cross-Entropy (BCE) loss with data augmentation and pseudo-labeling to improve robustness on synthetic 3D scene distributions.

- Applied BatchNorm-based test-time adaptation (TTA) and F1-optimized probability-space ensembling to improve classification performance under distribution shifts.

AI Chatbot with RAG, Tool Agents & Streaming UI 2026

- GitHub: pkd-prashant/ai-chatbot-rag-tools

- Built an agentic AI chatbot using LangGraph and LangChain with Groq-hosted Llama 3, featuring real-time token streaming

via a Streamlit frontend.

- Implemented Retrieval-Augmented Generation (RAG) over user-uploaded PDFs using FAISS vector store and Ollama

embeddings for context-aware, grounded document QA.

- Integrated external tool agents (calculator, Alpha Vantage stock API, DuckDuckGo web search) with LangGraph routing

decisions dynamically; added persistent multi-session memory via SQLite.

THESIS

Interpretability in Continual Learning Using Concept Bottleneck Models and Mitigating Catastrophic Forgetting

- Neural networks suffer from catastrophic forgetting because learning a new task changes shared parameters and can overwrite representations learned for previous tasks.
- Continual learning aims to learn new tasks sequentially while preserving the model's previously acquired knowledge.
- Proposed **SPADE**, a self-explainable class-incremental learning framework that learns a sparse and interpretable concept space from CLIP image embeddings using a Sparse Autoencoder(SAE), eliminating the need for manual prompt engineering and external language models.
- **Losses used:** Binary Cross-Entropy (BCE) loss, cosine similarity loss, reconstruction loss, and sparsity regularization loss.
- Achieved competitive/SOTA results on 6 coarse- and fine-grained CL benchmarks (CIFAR-100, Tiny-ImageNet, CUB-200, Food-101, Oxford Flowers-102, Stanford Cars), cutting trainable parameters by 28.6% vs. CLG-CBM.

PUBLICATIONS

SPADE: Sparse Prototype-Anchored Concept Bottleneck Model for Continual Learning submitted in AAAI 2027 (under review)

ACHIEVEMENTS

- Secured **AIR 72 among 28,000+ teams** in the **Amazon ML Challenge 2026**.

- Secured **3rd rank among 74 teams** in the IITH Deep Learning 2026 Hackathon.

- Secured **4th rank among 26 teams** in the KLA AI Hack x IIT-H 2026 competition.

- Finalist in **Smart India Hackathon 2022** as part of Team BITWISE NERDS.

- Developed an Android application for an NCERT and DoSEL problem statement.

- Solved over **1000 coding problems**, securing a **1700+ LeetCode rating**.

RELEVANT COURSEWORK

Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Foundations of Machine Learning, Deep Learning